BM4 Implicit Method

Physical state and one reduced Hairer projection per complete cycle

Explicit palindromic operations

All ABBA and BM4 presets inherit `PalindromicMethod`. For signed $d = ch$, the elementary operations are

$$A_d(u, v, t) = (u + df(t, v), v, t), \qquad B_d(u, v, t) = (u, v + df(t, u), t), \qquad T_d(u, v, t) = (u, v, t + d).$$

Only T advances time. A coefficient $c = 1/2$ means duration $h/2$; there is no additional half inside A or B. The optional exact C flow rotates the copy difference and requires an independent coupling object. Both BM4 presets omit C.

In chronological order, standard ABBA2 is $A_{h/2}, B_{h/2}, T_{h/2}, T_{h/2}, B_{h/2}, A_{h/2}$. The distinct symmetric variant $T_{h/2}, A_{h/2}, B_{h/2}, B_{h/2}, A_{h/2}, T_{h/2}$ evaluates every spatial shear at the midpoint time. Both exhibit second order on the tested nonautonomous rotation. Arbitrary palindromes do not inherit fourth or sixth order, and an arithmetic projection need not be reversible or symplectic.

`Composition.from_half` reflects a supplied half sequence. The generic `PalindromicImplicit` and `PalindromicMidpoint` methods select one exterior Hairer projection or a final arithmetic mean. The standard signed ABBA/BM4 recipes, grouped observations and passive momentum normalization remain unchanged. NumPy and JAX execute the same explicit operations and differentiate their retained shear inputs. See `docs/models/extended/simulation/palindromic-operations.md`.

Current uncoupled configuration

Both `BM4Implicit` and `BM4Midpoint` omit the harmonic coupling flow and no longer accept a coupling-frequency constructor argument. Their twelve coefficients, signed clock and projection placement are unchanged. The independent operation `HarmonicCoupling` retains C for future compositions. Existing observer/archive rates are fixed at zero for new runs; nonzero historical calculations require their original frozen implementation.

1 Twelve-stage BM4 base map

Let χ_h be a first-order map and $\chi_h^* = \chi_{-h}^{-1}$ its adjoint. One complete step uses the palindromic composition

$$\Psi_h = \chi_{b_{12}h} \circ \chi_{b_{11}h}^* \circ \cdots \circ \chi_{b_2 h} \circ \chi_{b_1 h}^*,$$

with alternating adjoint and direct maps in chronological order and

$$(b_1, \ldots, b_{12}) = (a_1, a_2, a_3, a_4, a_5, a_6, a_6, a_5, a_4, a_3, a_2, a_1),$$

$$\begin{aligned} a_1 &= 0.0792036964311957, & a_2 &= 0.1303114101821663, \\ a_3 &= 0.2228614958676077, & a_4 &= -0.3667132690474257, \\ a_5 &= 0.3246481886897062, & a_6 &= 0.1096884778767498. \end{aligned}$$

The half-cycle coefficients sum to $1/2$. Palindromy makes the composition self-adjoint and the order conditions give designed global order four. The negative coefficient a_4 is an intentional backward subflow.

2 Physical state and internal workspace

Twelve direct/adjoint stages enclosed by one spatial Hairer projection. Energy tracking is optional and passive. After convergence, passive energy quadrature evaluates $-\partial_t H$ at the 24 shear inputs retained from the accepted residual, with their original signed durations. It repeats no spatial map or coupling and performs no new nonlinear solve. Trial traces are discarded; energy evaluations remain outside physical solver counters. The common state conventions are specified below.

3 One Hairer projection around the complete cycle

For every complete BM4 step, the method displaces the diagonal embedding by a single multiplier $\mu \in \mathbb{R}^{2N}$,

$$\widehat{Y}_n = Ez_n + N\mu, \qquad M(\mu) = \Psi_{h,t_n}(\widehat{Y}_n), \tag{1}$$

$$r(\mu) = GM(\mu) + 2\mu, \qquad z_{n+1} = P\big(M(\mu) + N\mu\big). \tag{2}$$

The accepted multiplier is the root $r(\mu) = 0$. Thus the projection is applied once around the full twelve-stage composition, never between its internal stages.

With $J_{\mathrm{BM4}} = D\Psi_{h,t_n}(\widehat{Y}_n)$, Newton uses

$$D_\mu r = GJ_{\mathrm{BM4}}N + 2I.$$

The implementation forms J_{BM4} as the exact ordered product of the twelve guiding-centre stage Jacobians, with a centred-difference fallback. Good Broyden is retained as a nonlinear-solver choice; it changes the finite iteration path but not the defining projected map.

4 Fixed-grid lifecycle and diagnostics

The fixed controller advances accepted diagonal states on an output-independent grid. Off-grid samples use independent shadow steps. Only main steps contribute observations and nonlinear diagnostics: corrections, residual evaluations, residual norm, tolerance and multiplier norm. Coupling and Jacobian settings are recorded as metadata. Optional observers retain physical endpoints and twelve base-stage snapshots, allowing an independent energy reconstruction.

5 Implementation correspondence

The public class is in `src/methods/extended/bm4.py`. Its twelve-stage recipe is defined in `composition.py`, and its reduced equation uses `projection.py`. See `implicit-reduced.tex` for the projection derivation. The former BM4 import adapters have been removed.

The public names and exports from `simulation` remain unchanged. Both BM4 constructors now omit harmonic coupling. Internal imports use the defining modules; old import adapters have been removed. The ordinary implicit path is `advance`, `solve_projection`, `compose`, then explicit $A/B/T/C$ operations. Arithmetic midpoint variants use one explicit average instead of a nonlinear projection. Adjoint stages evaluate at the start of their signed substep; direct stages evaluate at its end.

Newton and Broyden share the spatial equations and convergence services. Exact particle-block tangents, passive energy and optional observation use retained shear inputs; energy is evaluated only after accepting the spatial map, with the existing factor 1/2 for physical κ. Time and momentum never enter the spatial root. The new internal family name does not enable the retired full time/momentum projection.

Current module responsibilities and diagrams are documented in `docs/models/extended/simulation/extended-simulation-architecture.md`.

6 Four state formulations and passive energy tracking

For one planar particle the supported internal representations are

physical	$z = (x, y)$	$\mathbb{R}^2$,
physical with energy	(z, t, κ)	$\mathbb{R}^4$,
duplicated	$(u, v), \quad u, v \in \mathbb{R}^2$	$\mathbb{R}^4$,
duplicated with energy	(u, v, t, κ)	$\mathbb{R}^6$.

Classical methods use the physical rows; ABBA and BM4 use the duplicated rows. `track_energy` selects the energy variant. For N planar particles the respective dimensions are $2N$, $4N$, $4N$, and $6N$: one time entry and one momentum per particle. All time entries equal the integration time. Classical full-cyclotron runs instead use $4N$ or $6N$ entries. Accepted spatial copies coincide; only one physical copy is exported. Every coordinate block contains N values in component-major order. The formulation validates particle times and aligns output round-off; the integrator does not interpret coordinate indices.

The normalized passive balance is

$$\dot{\kappa} = -\partial_t H(t, z), \qquad \kappa(t_0) = 0, \qquad \Delta K(t) = H(t, z(t)) + \kappa(t) - H(t_0, z_0).$$

Splitting accumulators from both copies are divided by two. The physical Hamiltonian may vary with time; ΔK measures its computed balance. Tracking leaves physical trajectories, nonlinear stopping rules and adaptive acceptance unchanged. Dynamics must supply the momentum derivative, including zero for an autonomous Hamiltonian.

Hairer's constraint is solely $u - v = 0$: the reduced multiplier has $2N$ entries and ABBA's simultaneous spatial solve has $6N$ unknowns. Neither t nor κ is projected. The former eight-component projection is retired; `state_extension='fully_extended'` raises migration guidance.

Dynamics, formulation, method and integration remain separate. The shared state classes own layouts, diagonal embedding and history extraction in

`src/formulations/state.py`

Each integration initializes a fresh method instance. Observers receive physical states; energy histories are in `Solution.diagnostics`. The `extended_time` history has shape (N, n_{samples}). Fixed-step shadow samples leave main states and counters unchanged. Adaptive energy quadrature uses accepted dense output; its backend state and Jacobian remain physical.

7 Optional JAX execution

An optional JAX driver executes the same numerical map on one selected CPU or GPU, with particle operations batched inside a compiled sequential time loop. It reuses the canonical coefficients and prepared potential splines. Physical-state tolerance scaling, signed stage times, passive energy weights and independent shortened output maps retain their definitions above. Applicable Newton systems use independent particle blocks; an explicitly selected Broyden iteration retains its global secant matrix. Nonlinear iteration limits and failure checks remain active on the device.

Both execution routes use double precision and return the same physical solution layout. Floating-point reassociation can produce round-off differences without changing the designed order or geometric properties. First-call compilation is separate from warmed integration timings, which include transfers and output construction. The sibling runtime document and shared execution contract specify supported callbacks and exact device scope.